HW4

Tanner Crane

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1 4.1 Matrix Algebra

1.1 7

To prove the formula for A^k given the block matrix $A = \begin{bmatrix} B & C \\ 0 & I \end{bmatrix}$ where $B - I$ is non singular, we can use mathematical induction on $k \geq 1$.

Base case: (k=1)

$$A^1 = \begin{bmatrix} B^1 & (B^1 - I)(B-I)^{-1}C \\ 0 & I \end{bmatrix} = \begin{bmatrix} B & (B-I)(B-I)^{-1}C \\ 0 & I \end{bmatrix} = \begin{bmatrix} B & IC \\ 0 & I \end{bmatrix} = \begin{bmatrix} B & C \\ 0 & I \end{bmatrix}$$

✓

Now assume that the formula holds for some integer $k \geq 1$, then

$$A^{k+1} = A^k A = \begin{bmatrix} B^k & (B^k - I)(B-I)^{-1}C \\ 0 & I \end{bmatrix} \begin{bmatrix} B & C \\ 0 & I \end{bmatrix}$$

$$(B^k)(B) + \left((B^k - I)(B-I)^{-1}C\right)(0) = B^{k+1}$$

$$(B^k)(C) + \left((B^k - I)(B-I)^{-1}C\right)(I) = \left[B^k + (B^k - I)(B-I)^{-1}\right]C$$

Let

$$B^k = B^k(B-I)(B-I)^{-1}$$

$$\begin{aligned} B^k + (B^k - I)(B-I)^{-1} &= \left[B^k(B-I) + (B^k - I)\right](B-I)^{-1} \\ &= \left[B^{k+1} - B^k + B^k - I\right](B-I)^{-1} \\ &= \left(B^{k+1} - I\right)(B-I)^{-1} \end{aligned}$$

Multiplying by C on the right yields the top-right block:

$$\left(B^{k+1} - I\right)(B-I)^{-1}C$$

Bottom-left block:

$$(0)(B) + (I)(0) = 0$$

Bottom-right block:

$$(0)(C) + (I)(I) = I$$

Combining gives $\rightarrow$

$$A^{k+1} = \begin{bmatrix} B^{k+1} & (B^{k+1} - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

By induction, the formula holds for all integers $k \geq 1$. $\square$

1.2 12

Let $M = \begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix}$. Then we need an inverse matrix of the form $M^{-1} = \begin{bmatrix} B & x \\ y^T & c \end{bmatrix}$, where B is an $n \times n$ matrix, x and y are vectors in $\mathbb{R}^n$, and c is a scalar. $MM^{-1} = I_{n+1}$:

$$\begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix} \begin{bmatrix} B & x \\ y^T & c \end{bmatrix} = \begin{bmatrix} I_n & 0 \\ 0 & 1 \end{bmatrix}$$

1.

$$AB + uy^T = I_n$$

2.

$$Ax + uc = 0$$

3.

$$v^T B = 0$$

4.

$$v^T x = 1$$

$$x = -cA^{-1}u$$

(Solving for x since A is invertible from equation 2)

Substitute into equation 4:

$$v^T(-cA^{-1}u) = 1 \implies -c(v^T A^{-1}u) = 1$$

For scalar c to exist,

$$v^T A^{-1}u \neq 0$$

This is the necessary and sufficient condition for vectors u and v.

Assume

$$v^T A^{-1} u \neq 0$$

and let k =

$$v^T A^{-1} u \implies c = -\frac{1}{k}, x = -(-\frac{1}{k}) A^{-1} u = \frac{1}{k} A^{-1} u$$

Using equation 1 find B:

$$B = A^{-1} - A^{-1} u y^T$$

Substitute this into equation 3:

$$v^T (A^{-1} - A^{-1} u y^T) = 0$$

$$v^T A^{-1} - (v^T A^{-1} u) y^T = 0$$

Since $v^T A^{-1} u = k$, we have:

$$v^T A^{-1} - k y^T = 0 \implies y^T = \frac{1}{k} v^T A^{-1}$$

Finally, substitute y^T back into our expression for B:

$$B = A^{-1} - \frac{1}{k} A^{-1} u v^T A^{-1}$$

To conclude, the matrix is invertible if and only if $v^T A^{-1} u \neq 0$. When this condition is met, let $k = v^T A^{-1} u$. The explicit formula for the inverse is:

$$\begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix}^{-1} = \begin{bmatrix} B & x \\ y^T & c \end{bmatrix} = \begin{bmatrix} A^{-1} - \frac{1}{k} A^{-1} u v^T A^{-1} & \frac{1}{k} A^{-1} u \\ \frac{1}{k} v^T A^{-1} & -\frac{1}{k} \end{bmatrix}$$

□

1.3 17

Let $x \in \mathbb{R}^n$ (Quadratic form)

$$x^T A x = (x^T A x)^T$$

then by the multiplicative property of the transpose:

$$(x^T A x)^T = x^T A^T (x^T)^T = x^T A^T x$$

Skew-symmetric means $A^T = -A$, so:

$$x^T A^T x = x^T (-A) x = -(x^T A x)$$

$$x^T A x = -(x^T A x)$$

$$2(x^T A x) = 0$$

$$x^T A x = 0$$

□

1.4 18

Let $A = [a_{ij}]$ be an $n \times n$ matrix over a field ($\mathbb{R}$ or $\mathbb{C}$). By definition of a skew-symmetric matrix, $A^T = -A$. Consider the relationship entry-wise, then:

$$a_{ji} = -a_{ij} \quad \forall i, j \in \{1, \ldots, n\}$$

Thus the diagonal elements are

$$a_{ii} = -a_{ii} \implies 2a_{ii} = 0 \implies a_{ii} = 0 \forall i.$$

□

Proving determinant is zero for odd order: (From definition $A^T = -A$.)

$$\det(A^T) = \det(-A)$$

By properties of the determinant ($\det(A^T) = \det(A)$ and $\det(cA) = c^n \det(A)$ for a scalar c)

$$\det(A) = (-1)^n \det(A)$$

Since n is odd, $(-1)^n$ = -1, thus

$$\det(A) = -\det(A) \implies 2\det(A) = 0$$

$$\det(A) = 0$$

. □

1.5 19

A_0 is symmetric by

$$A_0^T = \left[\frac{1}{2}(A + A^T)\right]^T = \frac{1}{2}(A^T + A) = A_0$$

A_1 is skew-symmetric by

$$A_1^T = \left[\frac{1}{2}(A - A^T)\right]^T = \frac{1}{2}(A^T - A) = -\frac{1}{2}(A - A^T) = -A_1$$

$$A_0 + A_1 = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T) = \frac{1}{2}(A + A^T + A - A^T) = \frac{1}{2}(2A) = A$$

We then substitute the decomposition $A = A_0 + A_1$ into the quadratic form yielding

$$x^T A x = x^T (A_0 + A_1) x = x^T A_0 x + x^T A_1 x$$

By properties of transposes (1x1 matrix is its own transpose) and skew-symmetry

$$x^T A_1 x = (x^T A_1 x)^T = x^T A_1^T x = x^T (-A_1) x = -x^T A_1 x \implies$$

$2x^T A_1 x = 0$, so $x^T A_1 x = 0$. Therefore:

$$x^T A x = x^T A_0 x + 0 = x^T A_0 x$$

□

2 4.2: LU and Cholesky Factorizations

2.1 1

(a) Let $X = U^{-1}$. Want to show $x_{ij} = 0$ for all $i > j$. Since U is upper triangular ($u_{ik} = 0$ for $k < i$) and invertible ($u_{ii} \neq 0$), the (i, j) entry of the product $UX = I$ for $i > j$ expands to:

$$0 = \sum_{k=i}^{n} u_{ik}x_{kj} = u_{ii}x_{ij} + \sum_{k=i+1}^{n} u_{ik}x_{kj}$$

For a fixed column j, solving this iteratively backward from row $i = n$ down to $j + 1$ shows that if $x_{kj} = 0$ for all $k > i$ the summation vanishes. This leaves $u_{ii}x_{ij} = 0$. Because $u_{ii} \neq 0$, $x_{ij} = 0$. By backward induction, $x_{ij} = 0$ for all $i > j$, proving U^{-1} is upper triangular. □

(b) Let $X = L^{-1}$. Since L is unit lower triangular ($l_{ii} = 1, l_{ik} = 0$ for $k > i$), the (i, j) entry of $LX = I$ isolates to:

$$x_{ij} = \delta_{ij} - \sum_{k=1}^{i-1} l_{ik}x_{kj}$$

For a fixed column j, evaluating iteratively forward from row $i = 1$ to j shows by induction that $x_{kj} = 0$ for all $k < i$, causing the summation to vanish. This yields $x_{ij} = 0$ for $i < j$, and $x_{jj} = 1$ for $i = j$. Thus, L^{-1} is unit lower triangular. □

(c) Let A and B be upper triangular. The (i, j) entry of AB is $c_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}$. Because $a_{ik} = 0$ for $k < i$ and $b_{kj} = 0$ for $k > j$, the terms in the summation are non-zero only when $i \leq k \leq j$. For $i > j$, this index range is empty, forcing the entire sum to vanish ($c_{ij} = 0$). Thus, AB is upper triangular. Reversing the index bounds ($i \geq k \geq j$) identically proves the lower triangular case. □

2.2 5

Let A be an $n \times n$ triangular matrix (either upper or lower). The determinant of a triangular matrix is the product of its diagonal entries

$$\det(A) = \prod_{i=1}^{n} a_{ii}$$

It is well known that A is nonsingular $\iff \det(A) \neq 0$.

$$A \text{ is nonsingular} \iff \prod_{i=1}^{n} a_{ii} \neq 0$$

By the zero-product property of fields (e.g., $\mathbb{R}$ or $\mathbb{C}$), a finite product of scalars is non-zero if and only if every individual factor is non-zero. Therefore:

$$\prod_{i=1}^{n} a_{ii} \neq 0 \iff a_{ii} \neq 0 \quad \forall i \in \{1, 2, \ldots, n\}$$

Thus, an upper or lower triangular matrix is nonsingular if and only if all of its diagonal elements are different from 0. $\square$